

MOTS-c

Mitochondrial-Derived Peptide

Metabolic Regulation · Exercise Mimetic · Longevity

What Is MOTS-c?

MOTS-c (Mitochondrial Open Reading Frame of the 12S rRNA type-c) is a 16-amino acid peptide encoded in mitochondrial DNA. It acts as a signaling molecule that regulates metabolism in response to stress, exercise, or fasting.

How It Works

Metabolic Activation Stimulates AMPK pathway to improve glucose uptake and fatty acid oxidation.	Exercise Mimetic Enhances mitochondrial function and energy production similar to physical activity.
Insulin Sensitivity Helps regulate blood sugar and combat insulin resistance.	Cellular Protection Supports antioxidant defenses and mitochondrial health under stress.

Areas of Research Interest

- Improved metabolic flexibility and fat burning
- Enhanced insulin sensitivity and glucose control
- Increased energy and physical endurance
- Support for healthy aging and longevity
- Potential benefits for obesity and diabetes research
- Mitochondrial function and cellular resilience

Common Research Approaches

Studied as an injectable peptide for metabolic and longevity research. Often combined with diet, exercise, and other interventions to mimic exercise benefits at the cellular level.

FOR RESEARCH PURPOSES ONLY · NOT FOR HUMAN CONSUMPTION

Educational only — not medical advice. Consult qualified professionals. All related compounds are not evaluated or approved as treatments for any condition.

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